

Is Tahoe Blue?

Assessing In Situ and Remotely Sensed Metrics of Water Clarity in Lake Tahoe

Introduction

Water clarity provides insight into the impacts of both natural phenomena and human activities on water bodies. Environmental monitoring and conservation efforts in the Lake Tahoe region have focused on water clarity for decades. Beyond its appeal to tourists and prospective residents alike, the exceptional clarity of Lake Tahoe's water also serves as a measure for the overall health of its ecosystem.

This study examines the clarity of Lake Tahoe's water with the use of both in situ (field) and remotely sensed measurements. While in situ measurements provide direct observations of environmental parameters at specific locations, satellite-derived metrics serve as proxies that can be related to these direct observations. Understanding the relationship between in situ observations and remotely sensed data is particularly valuable in environments where field observation is difficult.

Water clarity is assessed based on two distinct measurements: Secchi depth and the Normalized Difference Turbidity Index (NDTI). Secchi depth, a traditional field measurement, involves lowering a standardized white disk into water and recording the depth at which it is no longer visible, with greater depths indicating clearer water. On the other hand, NDTI is derived from satellite imagery and uses the differential reflectance of light in the red and green portions of the electromagnetic spectrum to detect suspended materials in water bodies; the index produces values ranging from -1 to 1, with lower values indicating clearer water and higher values indicating greater turbidity.

The primary objective of this study is to investigate the relationship between in situ Secchi depth measurements and satellite-derived NDTI values for Lake Tahoe over a ten-year period (2015-2024), focusing specifically on summer (August) observations when clarity issues are typically most pronounced. By analyzing the relationship between Secchi depth and NDTI, this study aims to assess the utility and/or limitations of NDTI as a supplementary monitoring tool for water clarity in Lake Tahoe.

Materials

A repository of historic Secchi depth readings for 1967 through 2021 is maintained by the Tahoe Environmental Research Center via the Environmental Data Initiative portal (2025). However, pre-processed data were provided as course materials for this study (UCLA GEOG 498, 2025), which contained averaged Secchi depth measurements for 2015 through 2024 in the month of August.

Secchi measurements were acquired at one of two monitoring stations: the LTP station (39.0972°, -120.1550°) and the Midlake station at (39.1417°, -120.0153°). Averaged Secchi depth measurements are available for all 10 years for LTP station; measurements from 2016 and 2024 are absent for the Midlake station.

Landsat 8 Collection 2 Level-2 surface reflectance imagery was acquired through the USGS EarthExplorer portal. Only scenes from Path/Row 043/033 were needed, for their extent exceeds that of Lake Tahoe proper. Scenes acquired in the month of August from 2015 to 2024 were selected based on the criteria that they exhibit less than 30% cloud cover.

This selection criteria yielded 18 usable Landsat scenes over the 10-year period. Years 2019 and 2022 yielded only one usable scene, but all other examined years yielded two satellite images captured in the month of August. In these cases, the NDTI metrics are averaged to produce a single annual observation.

Methods

Data processing of Secchi depth measurements and Landsat 8 imagery was performed in ArcGIS Pro with `arcpy`. Visualization was completed in ArcGIS Pro, while correlation analysis was completed in R.

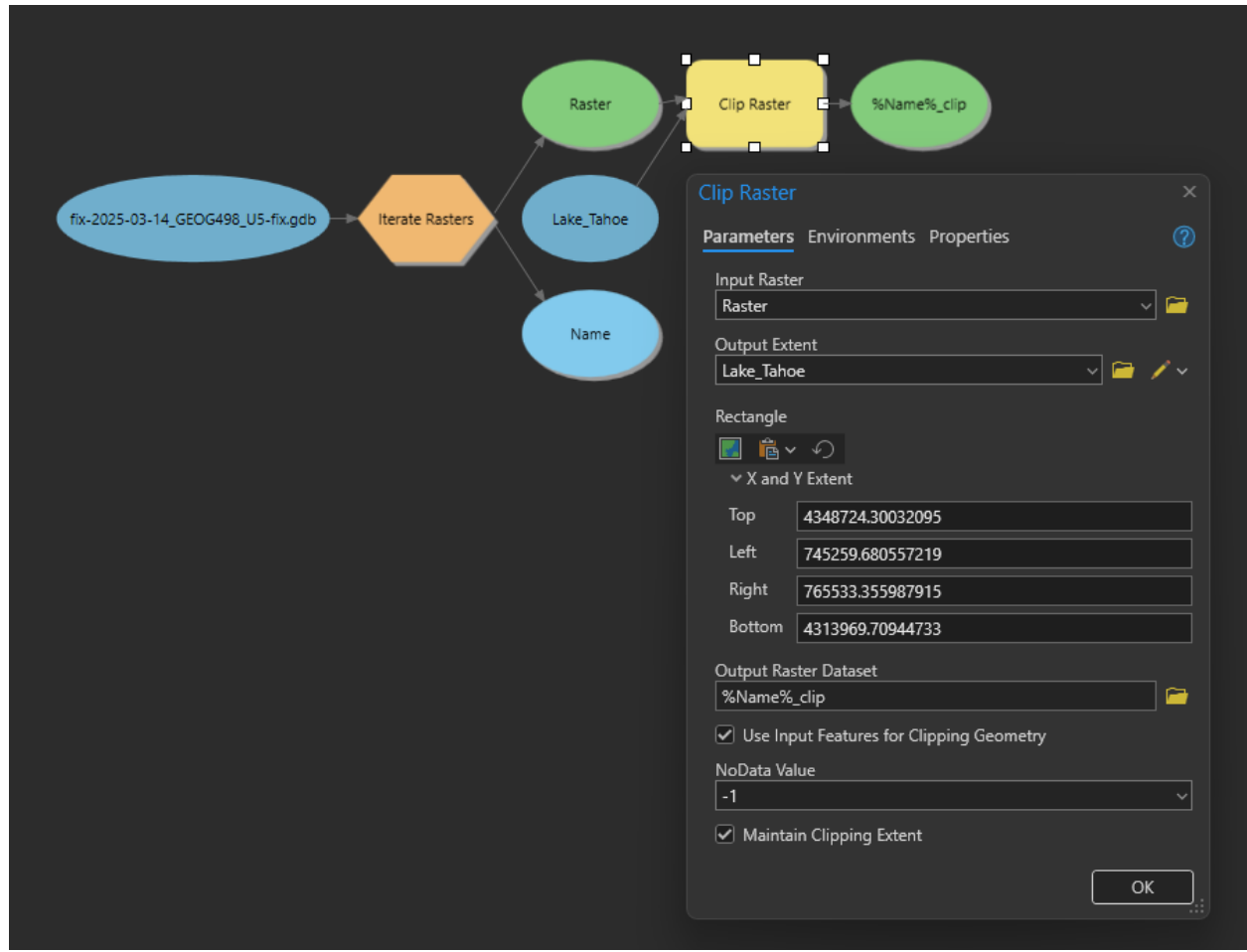
Historic averaged Secchi depth measurements for the LTP and Midlake monitoring stations were joined to point features for the stations, one-to-many, such that the “Secchi Station” point feature held a table of 18 rows, corresponding to 10 annual LTP station observations and 8 annual Midlake station observations. Two rows with null Secchi measurements were created for the Midlake station’s missing years: 2016 and 2024.

Landsat 8 imagery was first clipped to the shoreline boundary of Lake Tahoe. Next, a script was used to calculate $NDTI = ((Red - Green)/(Red + Green))$ for all 36 single-band raster images (see `script:NDTI-rasterCalculator.py`). A similar script was run to average the NDTI values for the 8 years where two images were acquired during the month of August (see `script:NDTI_avg-rasterCalculator.py`).

The averaged NDTI rasters for each year in the study period were then sampled at the Secchi depth monitoring stations, and the sampled values were joined to the “Secchi Station” table. This comprehensive table, showing Secchi measurements and NDTI samples for all 20 combinations of Year and Station, would be used by R to calculate the correlation coefficient and R^2 for the relationship between the two variables (see `script:correlation.r`).



modelBuilder:clipRaster





script:NDTI-rasterCalculator.py

```

'''{python}
import arcpy
import os
from arcpy.sa import *

# Check out
arcpy.CheckOutExtension("Spatial")

# Set input and output geodatabases
input_gdb = r"C:\Users\taket\Desktop\local\ArcGIS\2025-03-14_GEOG498_U5\fix-2025-03-14_GEOG498_U5-fix.gdb"
output_gdb = r"C:\Users\taket\Desktop\local\ArcGIS\2025-03-14_GEOG498_U5\fix-2025-03-14_GEOG498_U5-fix.gdb"

# Set workspace
arcpy.env.workspace = input_gdb

# List scenes
scene_ids = [
    "LC08_L2SP_043033_20150805_20200909_02_T1",
    "LC08_L2SP_043033_20150821_20200908_02_T1",
    "LC08_L2SP_043033_20160807_20200906_02_T1",
    "LC08_L2SP_043033_20160823_20200906_02_T1",
    "LC08_L2SP_043033_20170810_20201015_02_T1",
    "LC08_L2SP_043033_20170826_20200903_02_T1",
    "LC08_L2SP_043033_20180813_20200831_02_T1",
    "LC08_L2SP_043033_20180829_20200831_02_T1",
    "LC08_L2SP_043033_20190816_20200827_02_T1",
    "LC08_L2SP_043033_20200802_20200914_02_T1",
    "LC08_L2SP_043033_20200818_20200823_02_T1",
    "LC08_L2SP_043033_20210805_20210811_02_T1",
    "LC08_L2SP_043033_20210821_20210827_02_T1",
    "LC08_L2SP_043033_20220824_20220924_02_T1",
    "LC08_L2SP_043033_20230811_20230818_02_T1",
    "LC08_L2SP_043033_20230827_20230905_02_T1",
    "LC08_L2SP_043033_20240813_20240822_02_T1",
    "LC08_L2SP_043033_20240829_20240905_02_T1"
]

# Loop through each scene and calculate NDTI
for scene_id in scene_ids:
    print(f"Processing {scene_id}...")

    # Define input rasters
    red_raster = f"red_{scene_id}_SR_B4_clip"
    green_raster = f"green_{scene_id}_SR_B3_clip"

    # Define output rasters
    output_raster = os.path.join(output_gdb, f"NDTI_{scene_id}")

    # Calculate NDTI using Raster Calculator
    try:
        red = Raster(os.path.join(input_gdb, red_raster))
        green = Raster(os.path.join(input_gdb, green_raster))

        ndti = (red - green) / (red + green)

        # Save
        ndti.save(output_raster)
        print(f"Successfully created {output_raster}")

    except Exception as e:
        print(f"Error processing {scene_id}: {str(e)}")

# Check in
arcpy.CheckInExtension("Spatial")
print("Processing complete!")
'''

```



script:NDTI_avg-rasterCalculator.py

```

{python}
import arcpy
import os
from arcpy.sa import *

# Check out
arcpy.CheckOutExtension("Spatial")

# Set input and output geodatabases
input_gdb = r"C:\Users\taket\Desktop\local\ArcGIS\2025-03-14_GEOG498_U5\fix-2025-03-14_GEOG498_U5-fix.gdb"
output_gdb = r"C:\Users\taket\Desktop\local\ArcGIS\2025-03-14_GEOG498_U5\fix-2025-03-14_GEOG498_U5-fix.gdb"

# Set workspace
arcpy.env.workspace = input_gdb

# Years with multiple scenes (key=year, value=list of scene IDs)
year_scenes = {
    "2015": ["LC08_L2SP_043033_20150805_20200909_02_T1",
"LC08_L2SP_043033_20150821_20200908_02_T1"],
    "2016": ["LC08_L2SP_043033_20160807_20200906_02_T1",
"LC08_L2SP_043033_20160823_20200906_02_T1"],
    "2017": ["LC08_L2SP_043033_20170810_20201015_02_T1",
"LC08_L2SP_043033_20170826_20200903_02_T1"],
    "2018": ["LC08_L2SP_043033_20180813_20200831_02_T1",
"LC08_L2SP_043033_20180829_20200831_02_T1"],
    # "2019": only one scene, skip
    "2020": ["LC08_L2SP_043033_20200802_20200914_02_T1",
"LC08_L2SP_043033_20200818_20200823_02_T1"],
    "2021": ["LC08_L2SP_043033_20210805_20210811_02_T1",
"LC08_L2SP_043033_20210821_20210827_02_T1"],
    # "2022": only one scene, skip
    "2023": ["LC08_L2SP_043033_20230811_20230818_02_T1",
"LC08_L2SP_043033_20230827_20230905_02_T1"],
    "2024": ["LC08_L2SP_043033_20240813_20240822_02_T1",
"LC08_L2SP_043033_20240829_20240905_02_T1"]
}

# Process each year
for year, scenes in year_scenes.items():
    print(f"Processing {year}...")

    # Load the rasters for this year
    raster_paths = []
    for scene in scenes:
        raster_name = f"NDTI_{scene}"
        raster_path = Raster(os.path.join(input_gdb, raster_name))
        raster_paths.append(raster_path)

    # Calculate the average
    output_name = f"NDTI_{year}"
    output_path = os.path.join(output_gdb, output_name)

    # Add all rasters and divide by count
    ndti_avg = sum(raster_paths) / len(raster_paths)
    ndti_avg.save(output_path)

    print(f"Successfully created {output_path}")

# Check in
arcpy.CheckInExtension("Spatial")
print("Processing complete!")

```



script:correlation.r

```

{r}
setwd("F:/MAGIST/23_25w_GEOG_498_Capstone-I-GeospatialResearchMethods/U5_Spatial-Analysis-with-
Raster-Data-II/work")
library(tmap)
library(sf)
library(sp)
library(spdep)

secchiXndti <- read.csv("secchi-x-ndti.csv", sep=",", header=T)

correlation <- cor(secchiXndti$secchi_mean, secchiXndti$ndti_sample,
                  use="complete.obs", method="pearson")
Rsquared <- correlation^2
print("Landsat 8 NDTI vs. in situ Mean Secchi Depth (m)")
print(paste("Correlation coefficient:", correlation))
print(paste("R-squared value:", Rsquared))

# Filter data for complete observations
midlake_complete <- subset(secchiXndti, site=="Mid-Lake" & !is.na(secchi_mean) &
!is.na(ndti_sample))
ltp_complete <- subset(secchiXndti, site=="LTP" & !is.na(secchi_mean) & !is.na(ndti_sample))
complete_data <- secchiXndti[!is.na(secchiXndti$secchi_mean) & !is.na(secchiXndti$ndti_sample),]

# Create an empty plot with appropriate limits
plot(secchiXndti$ndti_sample, secchiXndti$secchi_mean,
     main="NDTI vs Mean Secchi Depth",
     xlab="NDTI (Normalized Difference Turbidity Index)",
     ylab="Mean Secchi Depth (m)",
     type="n", # "n" means no plotting, just set up the canvas
     xlim=range(secchiXndti$ndti_sample, na.rm=TRUE),
     ylim=range(secchiXndti$secchi_mean, na.rm=TRUE))

# Add points for each site with different colors
points(midlake_complete$ndti_sample, midlake_complete$secchi_mean, pch=19, col="blue")
points(ltp_complete$ndti_sample, ltp_complete$secchi_mean, pch=19, col="red")

# Add regression line
abline(lm(secchi_mean ~ ndti_sample, data=complete_data), col="black", lwd=2)

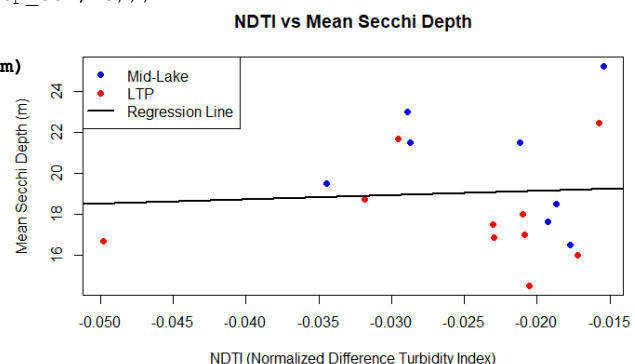
# Add a legend
legend("topleft",
      legend=c("Mid-Lake", "LTP", "Regression Line"),
      col=c("blue", "red", "black"),
      pch=c(19, 19, NA),
      lty=c(NA, NA, 1),
      lwd=c(NA, NA, 2))

# Calculate site-specific correlations
midlake_cor <- cor(midlake_complete$secchi_mean, midlake_complete$ndti_sample, method="pearson")
ltp_cor <- cor(ltp_complete$secchi_mean, ltp_complete$ndti_sample, method="pearson")

print(paste("Mid-Lake correlation coeff.:", round(midlake_cor, 3)))
print(paste("LTP correlation coeff.:", round(ltp_cor, 3)))

```

Landsat 8 NDTI vs. in situ Mean Secchi Depth (m)
Correlation coefficient: 0.0615790294767321
R-squared value: 0.00379197687129624
Mid-Lake correlation coeff.: -0.084
LTP correlation coeff.: 0.06



Results

The analysis of Secchi depth measurements and NDTI averages from 2015 to 2024 illuminates not only the temporal trends of Lake Tahoe's water clarity, but also the limitations of using remotely sensed data as a proxy for field observations. Figure 1 and Table 1 show Secchi Depth measurements for the LTP and Midlake monitoring stations. Figure 2 and Table 2 summarize the NDTI values calculated from Landsat imagery, sampled at both monitoring stations, along with the lake-wide average across the study period.

The most turbid conditions were observed in 2022, with NDTI values of -0.01576 and -0.015445 sampled at the LTP and Midlake stations, respectively (Figure/Table 2). The clearest water conditions were observed in 2021, with NDTI values of -0.049815 and -0.03448 sampled at LTP and Midlake stations. The lake-wide average ranged from -0.01781 (2022) to -0.03499 (2021), with an overall mean of -0.02407 across all years.

Secchi depth measurements showed considerable variability over the study period. Values measured at the LTP station ranged from 14.5 meters (2024) to 22.46 (2022), while those measured at the Midlake station exhibited an even wider range, from 16.5 meters (2023) to 25.25 meters (2022). Notably, 2022 exhibited both the highest Secchi depth readings (indicating clearer water) and the highest NDTI values (indicating more turbid water) at both stations. This contradiction is partially explained by correlation analysis (Figure/Table 3).

Surprisingly, correlation analysis between averaged Secchi depth measurements and averaged NDTI values yielded an extremely weak correlation coefficient ($r = 0.062$) with an R^2 value of 0.0038, indicating little to no linear relationship between the two metrics of water clarity. Site-specific analysis also revealed weak correlations, with the LTP station showing a weakly positive correlation ($r = 0.06$, $R^2 = 0.0036$) and the Midlake station showing a weakly negative correlation ($r = -0.084$, $R^2 = 0.0071$).

Year-to-year patterns show notable fluctuations in both metrics, with no clear trend emerging over the 10-year study period. The year 2022 stands out for having the greatest water clarity according to the Secchi depth metric, contradicted by the greatest turbidity asserted by the NDTI metric.

Figure 1

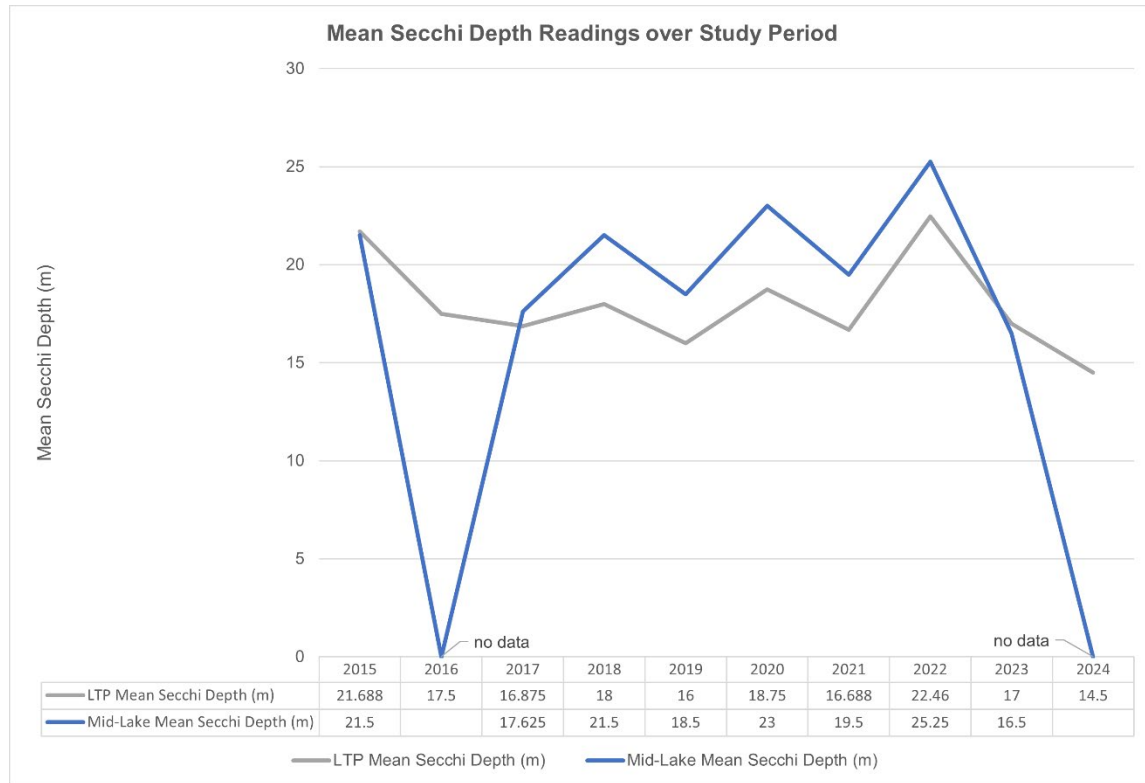


Table 1

Year	LTP Mean Secchi Depth (m)	Mid-Lake Mean Secchi Depth (m)
2015	21.688	21.5
2016	17.5	
2017	16.875	17.625
2018	18	21.5
2019	16	18.5
2020	18.75	23
2021	16.688	19.5
2022	22.46	25.25
2023	17	16.5
2024	14.5	

Figure 2

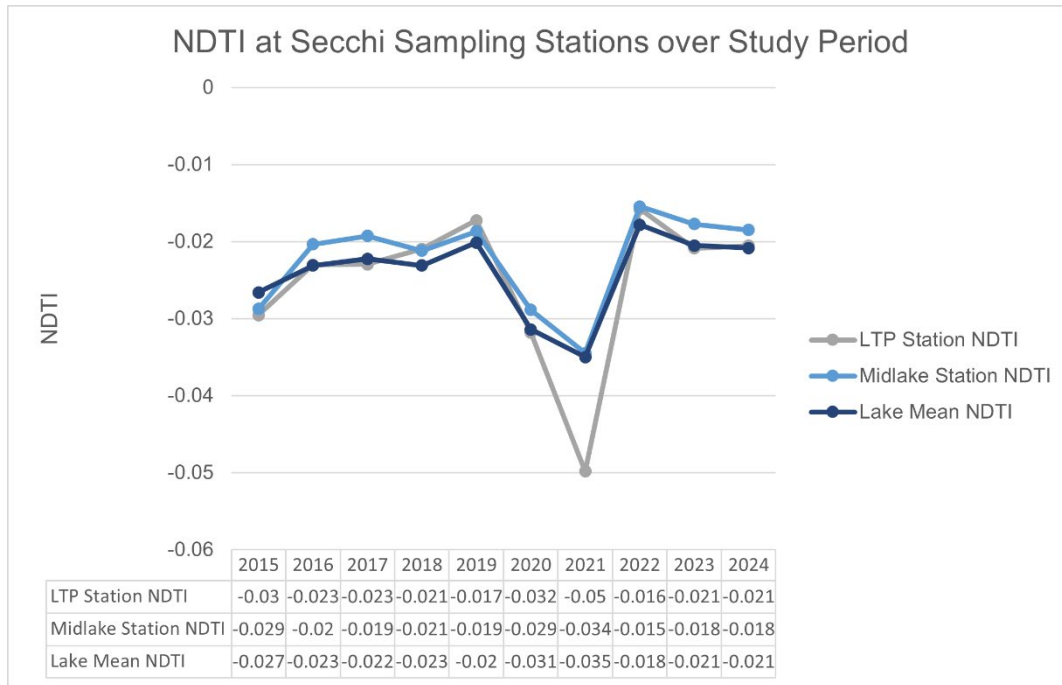


Table 2

Year	LTP Station NDTI	Midlake Station NDTI	Lake Mean NDTI	Lake Std. Dev. NDTI
2015	-0.029529	-0.028737	-0.0266	0.01172
2016	-0.023029	-0.020339	-0.02312	0.01127
2017	-0.022934	-0.01927	-0.02223	0.00827
2018	-0.020963	-0.021185	-0.02311	0.00893
2019	-0.017232	-0.018687	-0.02013	0.00948
2020	-0.031829	-0.028881	-0.03141	0.01077
2021	-0.049815	-0.034488	-0.03499	0.0116
2022	-0.01576	-0.015445	-0.01781	0.01172
2023	-0.020863	-0.01774	-0.02051	0.00884
2024	-0.020546	-0.018491	-0.02082	0.00977

Figure 3

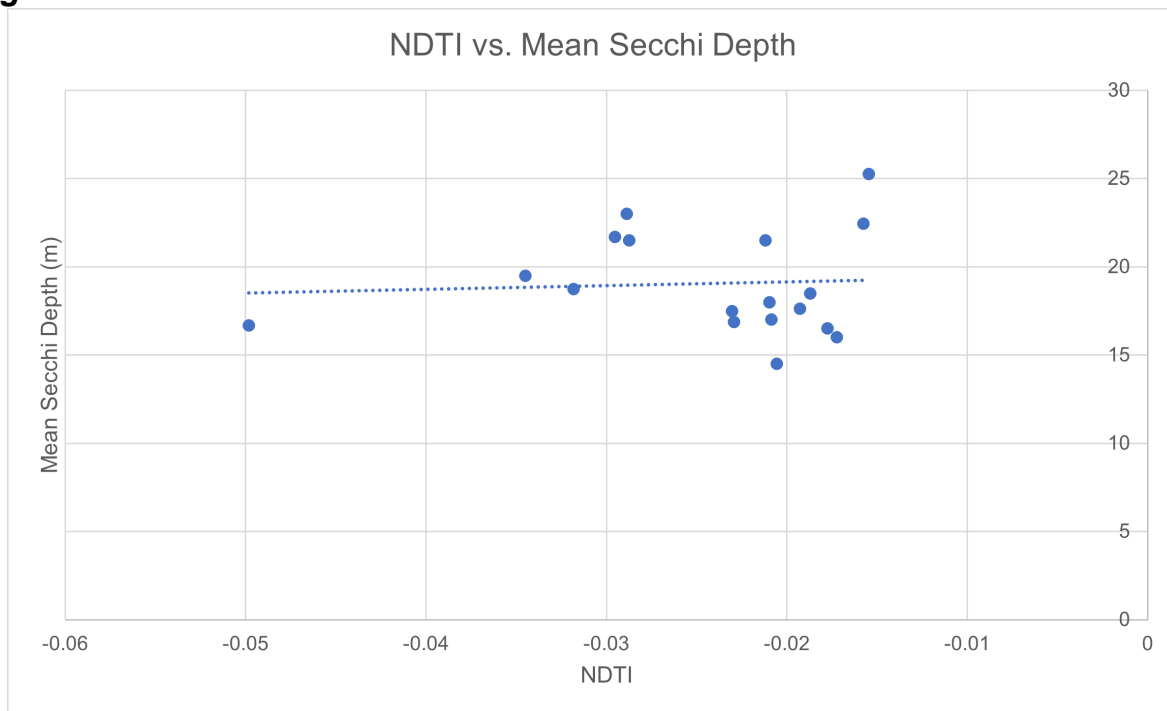
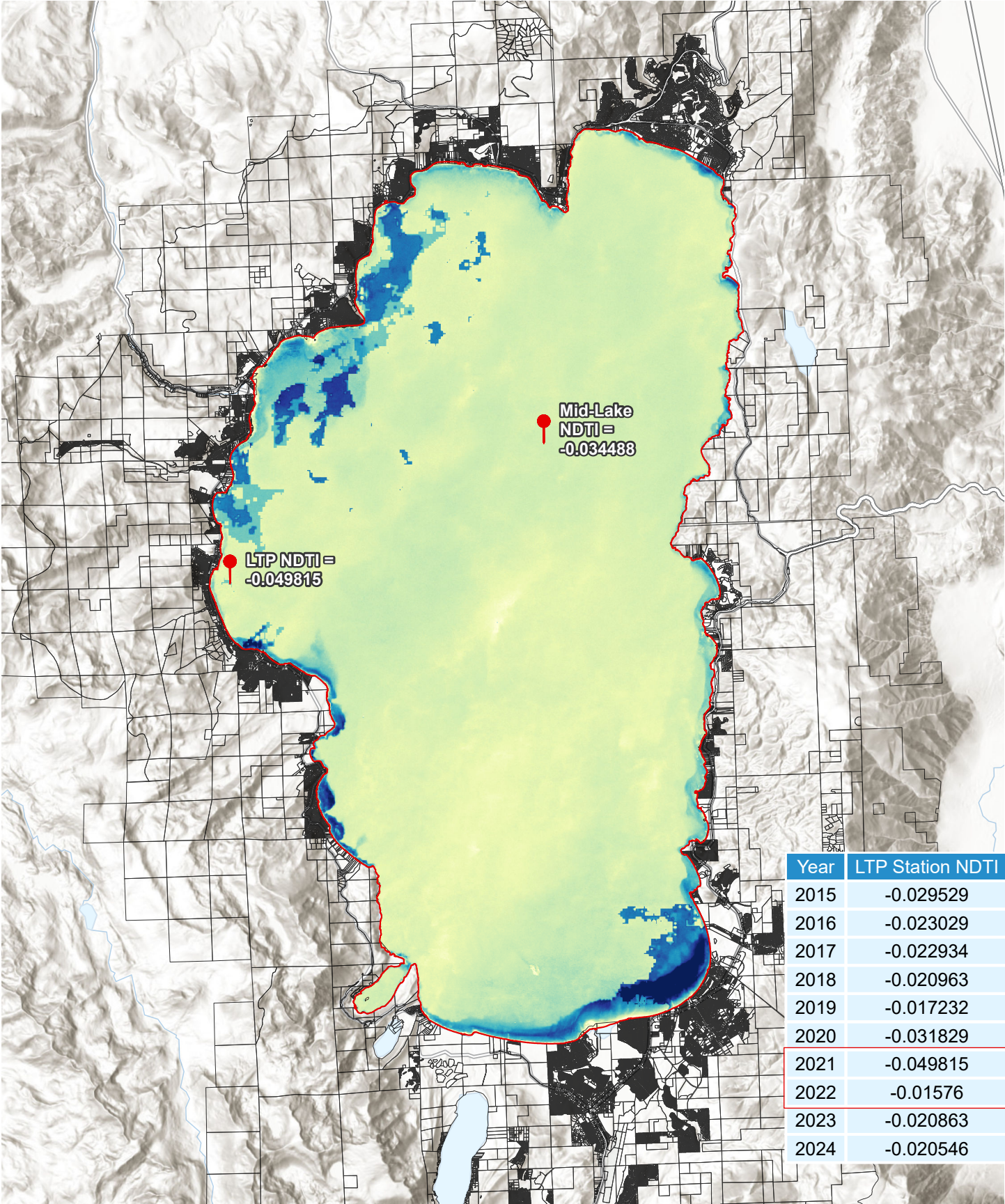


Table 3

Site	Year	Secchi Mean (m)	Secchi Std. Dev	Observations	NDTI	Latitude	Longitude
Mid-Lake	2015	21.5		1	-0.028737	39.1417	-120.0153
Mid-Lake	2016			0	-0.020339	39.1417	-120.0153
Mid-Lake	2017	17.625		1	-0.01927	39.1417	-120.0153
Mid-Lake	2018	21.5		1	-0.021185	39.1417	-120.0153
Mid-Lake	2019	18.5		1	-0.018687	39.1417	-120.0153
Mid-Lake	2020	23		1	-0.028881	39.1417	-120.0153
Mid-Lake	2021	19.5		1	-0.034488	39.1417	-120.0153
Mid-Lake	2022	25.25		1	-0.015445	39.1417	-120.0153
Mid-Lake	2023	16.5		1	-0.01774	39.1417	-120.0153
Mid-Lake	2024			0	-0.018491	39.1417	-120.0153
LTP	2015	21.688	0.088	2	-0.029529	39.0972	-120.155
LTP	2016	17.5	3.122	3	-0.023029	39.0972	-120.155
LTP	2017	16.875	0.629	4	-0.022934	39.0972	-120.155
LTP	2018	18	0.5	3	-0.020963	39.0972	-120.155
LTP	2019	16	1.732	3	-0.017232	39.0972	-120.155
LTP	2020	18.75	0.354	2	-0.031829	39.0972	-120.155
LTP	2021	16.688	2.23	4	-0.049815	39.0972	-120.155
LTP	2022	22.46	1.302	3	-0.01576	39.0972	-120.155
LTP	2023	17	0	2	-0.020863	39.0972	-120.155
LTP	2024	14.5	0	2	-0.020546	39.0972	-120.155

NDTI 2021

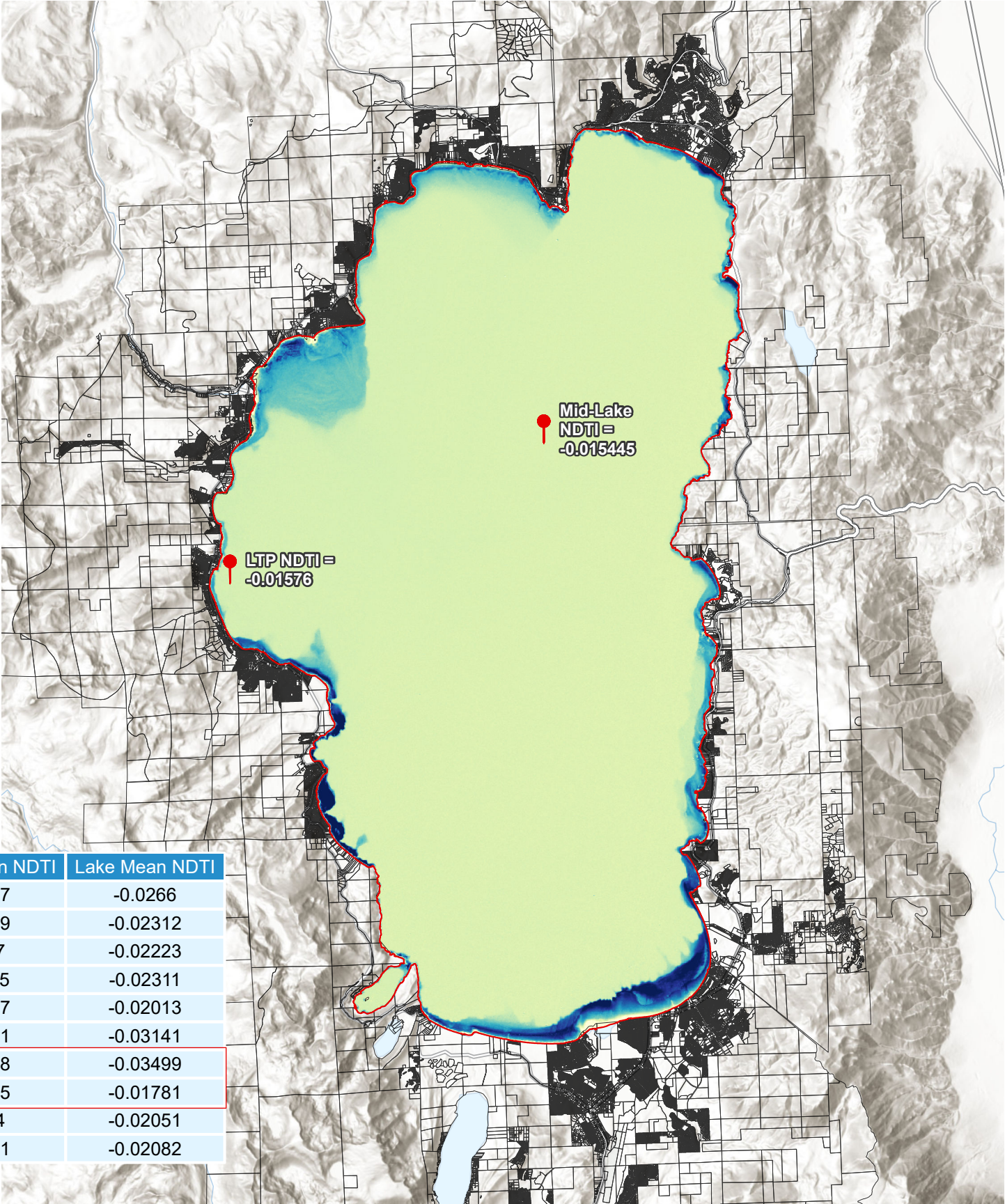
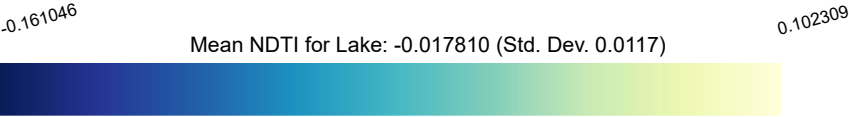
Lowest NDTI (clearest water)



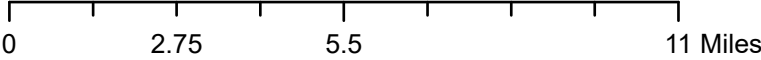
Year	LTP Station NDTI	Midlake Station NDTI	Lake Mean NDTI
2015	-0.029529	-0.028737	-0.0266
2016	-0.023029	-0.020339	-0.02312
2017	-0.022934	-0.01927	-0.02223
2018	-0.020963	-0.021185	-0.02311
2019	-0.017232	-0.018687	-0.02013
2020	-0.031829	-0.028881	-0.03141
2021	-0.049815	-0.034488	-0.03499
2022	-0.01576	-0.015445	-0.01781
2023	-0.020863	-0.01774	-0.02051
2024	-0.020546	-0.018491	-0.02082

NDTI 2022

Highest NDTI (greatest turbidity)



PCS: NAD 1983 UTM Zone 10N
GCS: GCS North American 1983
Projection: Transverse Mercator
1:200,000



Esri, NASA, NGA, USGS, Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, © OpenStreetMap contributors, and the GIS User Community

Data source: [Parcels \(TRPA\)](#), [Natural Rim \(TRPA\)](#), [Lake Tahoe Shoreline \(TRPA\)](#), [Secchi Depth \(TERC\)](#)
Cartography and analysis by Aaron Goodman for UCLA GEOG 498 Spring 2025
in report: pp. 12

Discussion

The weak correlation between Secchi depth and NDTI observed in this study challenges the utility of satellite-derived indices for monitoring water clarity in Lake Tahoe. Several factors could explain the discrepancies in findings between the in situ and remotely sensed measurements.

One challenge is the temporal alignment between satellite image acquisition and Secchi depth observations. While monthly averages were used for both metrics, day-to-day variations in water conditions could significantly impact the relationship between these measurements. Even with this study's imagery selection criteria, atmospheric conditions could introduce noise that affects the NDTI calculation. Some selected scenes did, in fact, contain notable smoke clouds from wildfires that obscured the satellite signal from the lake. Another challenge is that Secchi depth and NDTI are measuring different aspects of water clarity. NDTI may more heavily weigh surface-level conditions and the presence of specific materials like algae than Secchi depth's measurement of light penetration. The spatial scale difference between point-based Secchi readings and 30m-resolution Landsat 8 imagery may also introduce further discrepancies in the metrics. With each Landsat pixel corresponding to a 900 square meter area, finer-scale variations in water clarity may be lost which are otherwise captured by the Secchi measurements.

The counterintuitive results from 2022, where the Secchi depth metric indicates record clarity while the NDTI metric indicates record turbidity for the study period, highlight these challenges. This year exhibited conditions in which water clarity was excellent as observed by the Secchi metric, yet NDTI surface reflectance characteristics indicated higher turbidity.

Ultimately, this study is most limited by its examination of a limited time span, which only includes one summer month for 10 years, and features records with some missing data. Future studies could further the investigation into water clarity analysis for Lake Tahoe with higher temporal resolution sampling, additional water quality variables, and incorporation of alternative spectral indices with data of higher spatial and spectral resolution.

Works Cited

1. Watanabe, S. 2025. *Lake Tahoe Historic Secchi depth ver 11*. Environmental Data Initiative. doi.org/10.6073/pasta/f99b7c1b7ef7d7cb40c80f0f20134d58
2. UCLA GEOG 498. 2025. *Pre-processed Secchi Depth data* [Data set]. UCLA Geography. catalog.registrar.ucla.edu/course/2024/geog498
3. Tahoe Regional Planning Agency. 2021. *Parcels* [Data set]. TRPA GIS. maps.trpa.org/server/rest/services/DataDownloader_Parcels/MapServer
4. Tahoe Regional Planning Agency. 2021. *Shoreline_Contours* [Data set] TRPA GIS. maps.trpa.org/server/rest/services/Shoreline_Contours/MapServer

Landsat 8 Collection 2 Level-2 Scenes

[LC08 L2SP 043033 20150805 20200909 02 T1](#)
[LC08 L2SP 043033 20150821 20200908 02 T1](#)
[LC08 L2SP 043033 20160807 20200906 02 T1](#)
[LC08 L2SP 043033 20160823 20200906 02 T1](#)
[LC08 L2SP 043033 20170810 20201015 02 T1](#)
[LC08 L2SP 043033 20170826 20200903 02 T1](#)
[LC08 L2SP 043033 20180813 20200831 02 T1](#)
[LC08 L2SP 043033 20180829 20200831 02 T1](#)
[LC08 L2SP 043033 20190816 20200827 02 T1](#)
[LC08 L2SP 043033 20200802 20200914 02 T1](#)
[LC08 L2SP 043033 20200818 20200823 02 T1](#)
[LC08 L2SP 043033 20210805 20210811 02 T1](#)
[LC08 L2SP 043033 20210821 20210827 02 T1](#)
[LC08 L2SP 043033 20220824 20220924 02 T1](#)
[LC08 L2SP 043033 20230811 20230818 02 T1](#)
[LC08 L2SP 043033 20230827 20230905 02 T1](#)
[LC08 L2SP 043033 20240813 20240822 02 T1](#)
[LC08 L2SP 043033 20240829 20240905 02 T1](#)